JACKSON WILKE

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Data scientist and ML researcher with experience in large-scale climate modeling, civic data, and scientific computing.

EDUCATION

University of California, San Diego 2025 – Present
M.S. Data Science — Expected June 2027
Coursework: Introduction to Optimization, Statistical Models, Big Data Analysis with Dask

University of California, Los Angeles June 2024
B.S. Atmospheric & Oceanic Sciences and Mathematics

San Diego Mesa College June 2022
Certificate of Physics | Academic Honors Certificate

EXPERIENCE

Graduate Researcher — UCSD Climate Analytics Lab, San Diego, CA Sep 2025 – Present

- Developed ML pipeline predicting atmospheric variables from 60+ years of climate data (240×121 global grid) using PyTorch; built custom xarray/Dask data loaders for distributed processing of 800GB+ datasets.
- Built scalable preprocessing system handling missing data, normalization, spatial/temporal subsetting, and efficient batching.
- Engineered production ML training workflow with experiment tracking (Weights & Biases), hyperparameter management (Hydra), and automated checkpointing.
- Developing a coupled atmosphere–ocean model integrating JEM, a JAX-based physics atmospheric model, with Samudra, a PyTorch ML ocean emulator.
- Implemented a custom adapter layer and bilinear regridder to bridge incompatible grid systems between the two models.

Data Scientist — Hack for LA, Los Angeles, CA Aug 2024 – Sep 2025

- Contributed to an interactive web app mapping 12.5 million parking citations; analyzed large datasets to identify trends and actionable insights for the public.
- Collaborated with developers, engineers, and data scientists to deliver a cohesive product while continuously refining features.

Undergraduate Researcher — UCLA, Los Angeles, CA Mar 2023 – Sep 2023

- Applied Lagrangian particle tracking (OceanParcels) to analyze Southern Ocean currents contributing to ice shelf melting, using thickness-weighted velocity data from the MIT GCM (1/3° resolution).
- Processed tens of thousands of rows of 3D velocity data.

Undergraduate Researcher — Salk Institute for Biological Studies, San Diego, CA Sep 2021 – Sep 2022

- Analyzed olfactory data using hyperbolic geometry by mapping odors to points in 3D hyperbolic space.
- Applied Bayesian statistics (PyStan) and data visualization techniques to multi-dimensional datasets.
- Collaborated with graduate students, presenting research to professors and postdoctoral fellows.

SKILLS

Python (PyTorch, Xarray, Dask, Weights & Biases, Hydra, Pandas, Seaborn) • SQL • R • MATLAB

AWARDS

1st Place — Research Conference, San Diego Mesa College (2022): "Enjoying the Little Things in Life: A Study of Plankton in Mission Bay Emphasizing Microplankton", analyzing abundance and diversity of microplankton spatially and temporally.